

MilkWise Mobile — Data Storage & Sync Analysis

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The Question

How should the MilkWise mobile app store and sync data? Three user needs must be balanced:

1. **New phone continuity** — a parent upgrades their phone and doesn't lose feed history
 2. **Co-parent sync** — two parents in the same house see the same data in real time
 3. **Cost** — preferably free, or a natural upgrade upsell
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Option A — Device-Only (Local Storage)

How it works: All data lives in AsyncStorage on the phone. Nothing leaves the device.

Cost	Free forever
New phone	Data lost unless manually exported (CSV) and re-imported
Co-parent sync	Not possible
Privacy	Best — data never leaves the device
Complexity	Zero — already built

Verdict: Fine for solo use, but losing data on a new phone is a dealbreaker for most parents. Not viable as the only option.

Option B — iCloud / Google Drive (Native Platform Sync)

How it works: Use [expo-file-system](#) + iCloud Documents (iOS) or Google Drive App Data (Android) to sync the JSON data file automatically.

Cost	Free — uses the user's own iCloud/Google storage quota
New phone	Seamless — restores automatically on new device
Co-parent sync	⚠ Limited — iCloud syncs between the same Apple ID's devices; different accounts (two iPhones) won't share data
Privacy	Good — stored in user's own cloud account
Complexity	Medium — platform-specific implementation; iOS and Android behave differently

Verdict: Excellent for single-user continuity across devices. Co-parent sync only works if both parents use the same Apple ID — unusual and impractical. Not a full solution.

Option C — Supabase (Free Tier)

How it works: Supabase is an open-source Firebase alternative with a generous free tier. Each user gets an account; feeds are stored in a Postgres database in the cloud. Co-parents join the same "household" via a short join code and see shared data in real time.

Cost	Free up to 500MB DB, 50,000 monthly active users. Paid from \$25/month for larger scale
New phone	Seamless — log in on new device, data appears
Co-parent sync	Full — any device in the same household sees the same data in real time
Privacy	⚠ Data on Supabase servers (EU region available)
Complexity	Medium — requires auth flow (email/password or magic link), household concept
Free tier ceiling	50,000 MAU is very generous — MilkWise could run free for years at typical growth

Verdict: Best overall option for co-parent sync. The free tier is large enough to run the entire app for free at early scale. Risk: if Supabase changes pricing, costs could emerge — but data can be self-hosted on a VPS if needed.

Household design (Supabase)

Data model:

```

Household
  id          uuid  PRIMARY KEY
  join_code   text  UNIQUE    -- e.g. "MW-4X9K", shown in Settings
  created_at  timestampz

Profile
  id          uuid  PRIMARY KEY  -- Supabase Auth user id
  household_id uuid  REFERENCES household(id)
  display_name text

Feed
  id          uuid  PRIMARY KEY  -- generated on device
  household_id uuid  REFERENCES household(id)
  timestamp   bigint                -- Unix ms
  volume      numeric
  updated_at   timestampz           -- for conflict resolution
  deleted_at   timestampz           -- soft delete; NULL = active

```

Join flow (designed for exhausted parents): 1. First parent taps "Enable sync" in Settings → account created (email + password or magic link) → household auto-created → 6-character join code shown (e.g. **MW-4X9K**) 2. Second parent installs app → taps "Join household" → enters code → instantly linked 3. Both see the same feeds in real time via **Supabase Realtime** (WebSocket subscription on **feed** table filtered by **household_id**)

Offline behaviour: - Feeds always log to AsyncStorage first (instant, no network required) - Background sync to Supabase when connection is available - Conflict resolution: last-write-wins per feed UUID (safe — feeds are mostly append-only; edits and deletes propagate via **updated_at** / **deleted_at**) - On reconnect: client pushes any locally-queued writes, then pulls any remote changes since last sync timestamp

What Pro gates: - Free: local-only + iCloud/Google Drive backup (no account, no friction) - Pro: Supabase sync enabled, join code available, co-parent invite flow

Upsell message:

"Two parents, one view. Enable household sync and anyone at home can log feeds and see the same data — in real time, on any phone."

Option D — Self-Hosted API (The IDEA Pi approach)

How it works: The web app's API (already built, running on the Pi) is extended to accept mobile connections. Data stored on the Pi. This is essentially what the web app already does.

Cost	Free — uses the Pi the user already has
New phone	Seamless
Co-parent sync	Full
Privacy	Best — on-premises
Complexity	High — requires the user to have the web app running and reachable (Tailscale or port forwarding)

Verdict: Perfect for IDEA-internal use and power users. Not viable as the default for a public app — too much setup required.

Option E — Firebase (Google)

How it works: Same concept as Supabase but Google's product. Firestore for real-time sync.

Cost	Free tier: 1GB storage, 50,000 reads/day, 20,000 writes/day. Generous for feeding data (small payloads)
New phone	
Co-parent sync	Real-time via Firestore listeners
Privacy	⚠ Google
Complexity	Medium

Verdict: Functionally equivalent to Supabase. Firebase has better React Native SDKs but locks data into Google infrastructure. Supabase is preferred for independence.

Recommendation

Tier 1 — Free (Default)

Local storage + iCloud/Google Drive backup

- Data stored on device
- Auto-backup to the user's own cloud account (iCloud for iPhone, Google Drive for Android)
- Restores on new phone automatically
- Zero cost to you, zero cost to user
- No account required

Tier 2 — Pro (€4.99 one-time or included with Pro)

Supabase household sync

- Requires creating a MilkWise account (email + password)
- Creates a "household" — share a code with co-parent to join
- Real-time sync: both parents see feeds, both can log
- Works across any device, any platform
- Supabase free tier covers this indefinitely at early scale

Cost to you: - Supabase free tier: \$0/month up to 50,000 MAU - If MilkWise ever reaches 50,000 active households syncing: \$25/month — at which point revenue from Pro upgrades would far exceed this

Upsell message for users:

"Two parents, one view. Enable household sync and any parent at home can log feeds and see the same data — in real time, on any phone."

Summary Table

Option	New phone	Co-parent sync	Cost to you	Cost to user
A — Local only			\$0	\$0
B — iCloud/GDrive		⚠ Same account only	\$0	\$0
C — Supabase			\$0 (free tier) → \$25/mo at scale	\$0 (included in Pro)
D — Self-hosted			\$0 (Pi)	Setup effort

Option	New phone	Co-parent sync	Cost to you	Cost to user
E — Firebase			\$0 (free tier)	\$0

Proposed architecture: - **Free tier:** Local + iCloud/Google Drive (no account, no friction) - **Pro tier:** Supabase household sync (account required, upsell opportunity)